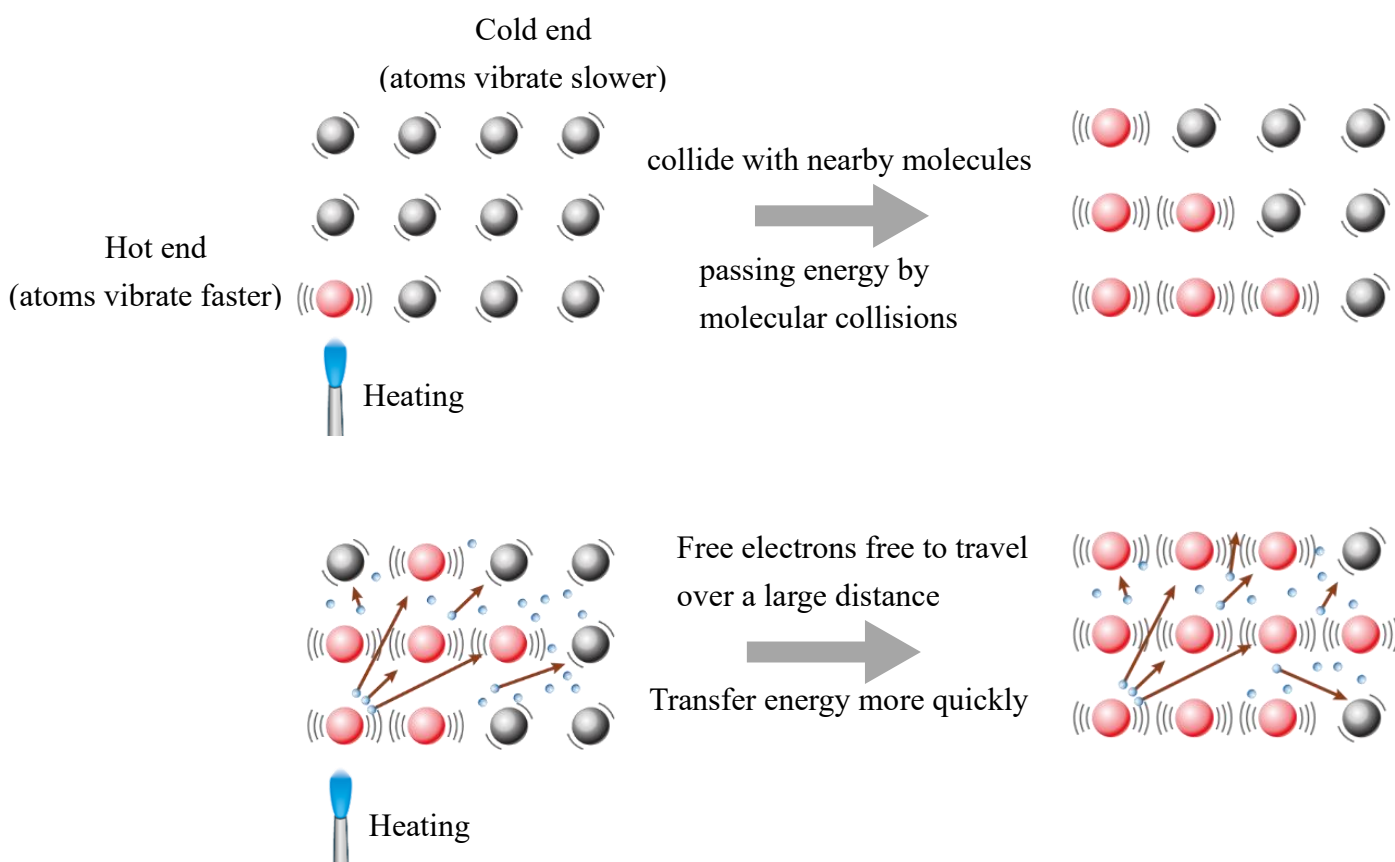


Heat transfer processes

- * **Conduction**
- * **Convection**
- * **Radiation**

(i) **Conduction**

- (1) Molecules in a solid are **closely packed** in regular structure. They vibrate but **can't move freely**.
- (2) When one end of a solid is heated, the molecules **gain kinetic energy** and vibrate **more violently**.
- (3) The heated molecules then collide with adjacent molecules and transfer energy to them. **Heat** is then **conducted** through the solid.



Remarks:

- **Conduction** takes place between two bodies **in contact**.
- **Conduction** can occur for solids, liquids or gases but **not in vacuum**.
- Different materials conduct heat at different rate. (solid > liquid > gas)
- Among solids, metal is a **good conductor (poor insulator)**. Plastic, glass and wood are **poor conductors (good insulators)**.

Example of conduction and insulation

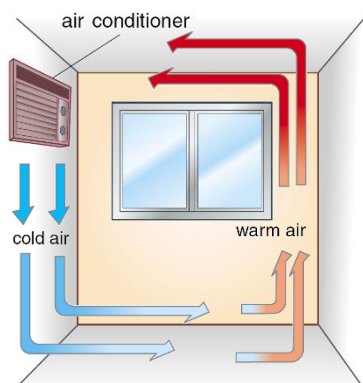
Enhancing conduction	Reducing conduction
(a) Frying pan - enhancing heat gain by the food - made of metal (good conductor) so that heat is conducted to the food quickly	(b) Oven glove - reducing heat gain by the hand wearing - consists of heat-insulating material (poor conductor) so that heat is conducted to the hand slowly
(c) Heat sink - enhancing heat loss of the processor - made of metal (good conductor) so that heat is conducted to air quickly to cool down the processor behind	(d) Polar bear's hair - reducing heat loss of the bear's body - air (poor conductor) is trapped between and inside the hair so that heat is conducted away from the body slowly to keep the bear warm

(ii) Convection

Convection is the process of heat transfer through a fluid (a liquid or a gas) by the movement of the fluid itself.

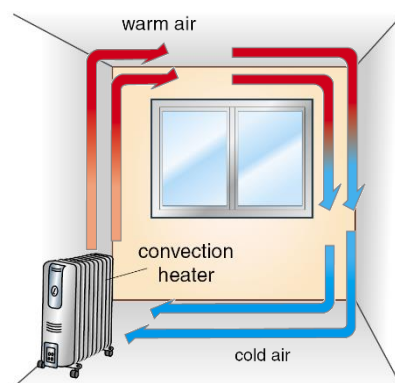
(1) Air conditioner:

- Air conditioner gives out cold air, which falls and pushes up the warm air to set up a convection current.
- Air conditioner usually set up high on the wall



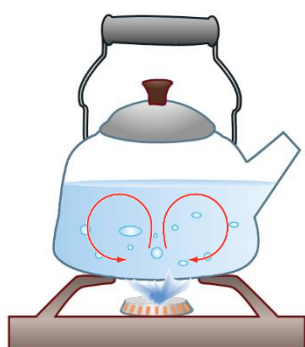
(2) Convection heaters:

- Air is warmed by conduction near the heater.
- Hot air rises and cold air sinks.
- Convection currents are set up.
- Convection heater is usually placed on the floor.



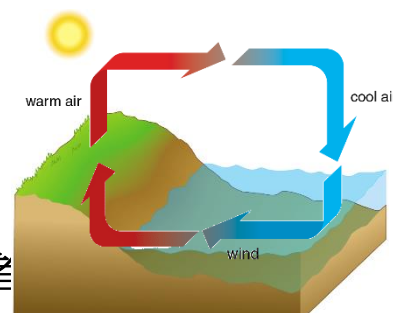
(3) Heater in electric kettle:

- When water is heated, water near the heater expands and become less dense than other region.
- Hot water rises and cold water sinks.
- Cooler water move in to replace the hot water.



(4) Sea-land breezes:

- During the day, the sun heats up the land faster than the sea.
- Warm air above the land rises and cold air blows in from the sea to replace warm air, then sea breezes resulted.



(iii) Radiation

- All hot or warm objects emit **infra-red radiation**. The hotter the object, the more infra-red radiations are emitted.
- Infra-red radiation can travel through vacuum.
- **Absorption** of infra-red radiation:
Black and **dull** surfaces are good **absorbers** of radiation.
White and **shiny** surfaces are poor **absorbers** of radiation
- Emission of infra-red radiation:
Black and **dull** surfaces are good **radiators** of radiation.
White and **shiny** surfaces are poor **radiators** of radiation



dark or dull
black surface



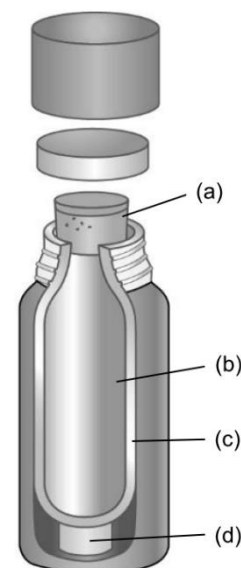
light and
shiny surface

Examples with multiple ways of controlling heat transfer

(1) Vacuum flask:

A vacuum flask **keeps** drink **hot and cold** at a **stable temperature**.
It has different components which reduce heat transfer by conduction, convection and radiation.

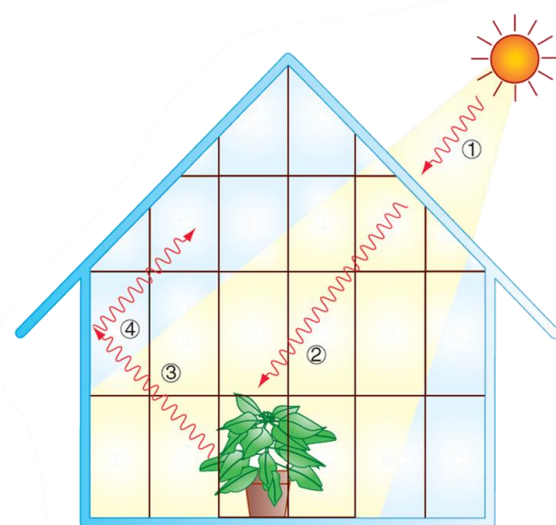
- Plastic or cork stopper
reduces heat transfer by **conduction** and **convection**
- Silvery glass or stainless steel walls
reduce heat transfer by **radiation**
- Vacuum between double walls
prevent heat transfer by **conduction** and **convection**
- Insulated support
reduce heat transfer by **conduction**



(2) Greenhouse:

A greenhouse provides a warm environment for growing plants.

- The **radiation** from the sun enters the greenhouse.
- The contents inside greenhouse **absorb** the radiation and become warm.
- Warm objects in turn emit radiation, mostly in form of **infra-red radiation**.
- The infra-red radiation **cannot pass through** the glass and is **trapped inside**.



Differences among conduction, convection and radiation

	<i>Conduction</i>	<i>Convection</i>	<i>Radiation</i>
Temp. difference	Presence	Presence	Absence
Medium	Material is needed	Fluid is needed	No medium is needed
Mobility of matter	No material is transferred	The fluid medium moves about	No material is transferred
Factors affecting rate of energy transfer	● Temp. differences ● Nature of material	● Temp. differences ● Nature of fluids	● Temp. of object ● Nature of surface
Speed of transmission	Very slow	Slow	The fastest

1. In which of the following media can conduction take place?

(1) Metal

(2) Glass

(3) Water

A (1) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

D

Conduction takes place in all kinds of media.
∴ (1), (2) and (3) are correct.

2. Which of the following transfer processes of heat can take place in a hot fluid?

(1) Conduction

(2) Convection

(3) Radiation

A (1) only

B (3) only

C (1) and (2) only

D (1), (2) and (3)

D

Conduction takes place in all kinds of media.

∴ (1) is correct.

Convection takes place in fluids only.

∴ (2) is correct.

Radiation takes place with or without a medium as long as the medium is not opaque to the radiation.

∴ (3) is correct.

3. What kind of means of energy transfer does a fan used to help to reduce heat?

I. Conduction

II. Convection

III. Radiation

A. I only

B. II only

C. III only

D. I and II only

B

4. How was the heat energy from the sun transferred to the earth?

- I. Conduction
- II. Convection
- III. Radiation

- A. I only
- B. III only
- C. II and III only
- D. I, II and III

B

5. Which of the following statements is/are correct?

- (1) Conduction can take place in any medium except a vacuum.
- (2) Convection can take place in any medium except a vacuum.
- (3) Radiation can only take place in a vacuum.

- A (1) only
- B (2) only
- C (1) and (2) only
- D (2) and (3) only

(1) is correct.

Convection takes place in fluids only.

∴ (2) is incorrect.

Radiation takes place with or without a medium as long as the medium is not opaque to the radiation.

∴ (3) is incorrect

A

6. Which of the following does **not** affect the rate of conduction?

- A Cross-sectional area of the conductor
- B Materials which make up the conductor
- C Orientation of the conductor
- D Temperature difference between the two ends of the conductor

C

7. Which of the following statements about radiation is/are correct?

- (1) Dull black surfaces can slow down the rate of heat loss by radiation.
- (2) Dull black surfaces are good absorbers of radiation.
- (3) Shiny surfaces do not absorb radiation.

- A (1) only
- B (2) only
- C (1) and (3) only
- D (2) and (3) only

Dull black surfaces are good absorbers and good radiators of radiation. It does not slow down the rate of heat loss by radiation.

∴ (2) is correct and (1) is incorrect.

Shiny surfaces are poor absorbers of radiation. They absorb less radiation than dull black surfaces.

∴ (3) is incorrect.

B

8. Which of the following statements about radiation is/are correct?

- (1) An object at 0°C does not emit radiation.
- (2) An object at 2000°C absorbs radiation.
- (3) An object at the same temperature as the surroundings does not emit radiation.

- A (2) only
- B (1) and (3) only
- C (2) and (3) only
- D (1), (2) and (3)

B

All objects at all temperatures emit radiation and absorb radiation at the same time.
 $\therefore$ (1) and (3) are incorrect, (2) is correct.

9. Which of the following statements about heat is / are true?

- (1) Heat is used to describe the total energy stored in a body.
- (2) Heat is used to describe the energy transferred from one body to another as a result of a temperature difference between them.
- (3) A body's internal energy is increased when it is heated.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

D

10. If there is no net heat flow from one body to another when they are in thermal contact, then the two bodies must have the same

- A. Temperature.
- B. Kinetic energy.
- C. Potential energy.
- D. Internal energy.

A

11. The temperature of an object increase from 250 K to 300 K . Find the temperature change in degree celsius.

- A. -223°C
- B. -173°C
- C. 50°C
- D. 323°C

C

12. The temperature of an object is 20°C and that of another object is 280 K . Find the temperature difference of the two objects.

- A. 20 K
- B. 260 K
- C. 13°C
- D. 533°C

C

13. Name the three methods by which thermal energy is transferred.

The three methods are: conduction, convection and radiation.

14. (i) Why are white clothes recommended for summer?
(ii) By what process is the energy transfer from the sun to the earth?

- (a) It is because white clothes are poor absorbers of infrared radiation.
(b) Radiation

15. When you open a refrigerator door, why do your feet first feel the cool air?

It is because cold air has a higher density than air at room temperature. Consequently, the cold air sinks while the warm, air rises.

16. The figure shows a domestic electric kettle. It consists of an immersion heater.
Its surface is chromed silvery.

- (a) Which processes, conduction, convection and radiation, are involved in heating water by the heater? (2 marks)
(b) Explain briefly why the surface of the kettle is chromed silvery. (1 mark)



(a) Heat flows from the heater to the water close to the heater by conduction. It is then carried to other part of water by convection. (1)+(1)

(b) It is to reduce the heat loss by radiation. (1)

17. The figure below shows two ways to hold sausages by a barbecue fork. When the barbecue fork is put over a fire, which way can cook the sausages more evenly? Explain briefly. (3 Marks)

Way A can cook the sausages more evenly.

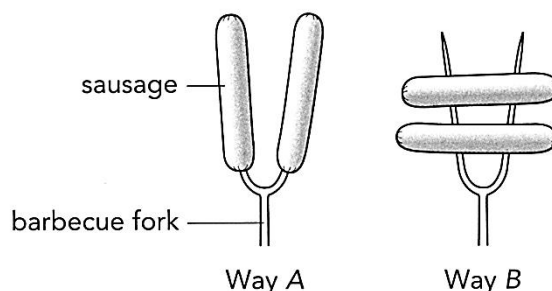
(1)

This is because the contact surface area between the sausages and the barbecue fork is larger.

(1)

Therefore, heat can be transferred (conducted) to the inside of the sausages more easily.

(1)



18. In a barbecue, Jenny suggests cooking some potatoes by wrapping the potatoes in aluminium Foil and putting them next to the charcoal fire

- Briefly explain the main processes that transfer heat from the fire to the potatoes. (2 marks)
- Explain why the potatoes should be wrapped in aluminium foil. (2 marks)
- One side of the aluminium foil used is shiny while the other side is dull. Which side should be facing outward? Explain briefly. (2 marks)

(a) Heat is transferred from the fire to the aluminium foil mainly by radiation.

(1)

Then, heat is transferred from the foil to the potatoes by conduction.

(1)

(b) The aluminium foil reflects the radiation falling on the potatoes and thus prevents overcooking.

(1)+(1)

(c) The shiny side should be facing outward because it is a better reflector of radiation.

(1)+(1)

19. (a) Consider a tile floor and a wooden floor both at the room temperature of 20 °C.
- We feel differently when walking barefoot on the two types of floor. Which one feels cooler? Explain briefly. (4 marks)
 - Two identical ice cubes are put respectively on the tile floor and the wooden floor. What will happen? Explain briefly. (3 marks)
- (b) A defrost tray is a metal plate that can defrost frozen food on it quickly. No heat source or electricity is needed. Explain its working principle base on the phenomenon discussed in (a). (2 marks)

- (a) (i) We feel the tile floor cooler than the wooden floor.

The floor is colder than our feet, so heat is transferred from our feet to the floor.

Tile is a better conductor of heat than wood, so it conducts the heat away from the feet faster.

- (ii) The floor is hotter than the ice cubes, so heat is transferred from the floor to the ice cubes.

Tile is a better conductor of heat than wood, so it conducts heat to the ice cubes faster.

As a result, the ice cube on the tile floor melts before the one on the wooden floor.

- (b) The plate is a very good heat conductor.

It conducts heat to the frozen food quickly to defrost the food.

20. The figure below shows a vacuum flask. It is commonly used to store hot liquids.

(a) Briefly describe how the flask prevents energy loss by conduction, convection and radiation.

(4 marks)

(b) Can the flask be used to keep a cold liquid cold? Explain briefly.

(2 marks)

(a) The flask consists of double walls. The vacuum layer between the double walls prevents energy loss by conduction and convection.

The inner wall is made of silvery glass or stainless steel which reduces energy loss by radiation.

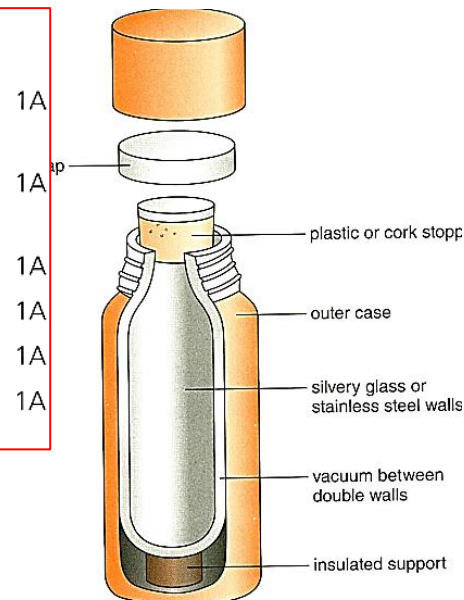
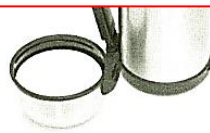
The plastic or cork stopper reduces energy loss by conduction and convection.

The insulated support at the bottom reduces energy loss by conduction.

(b) Yes,
the flask reduces the heat transfer between the contents and the surroundings.

MISCONCEPTION

Some students do not understand what a vacuum is. They think that there is still a little conduction and convection between the double walls.



21. In summer, we often cool down a room by air-conditioners.

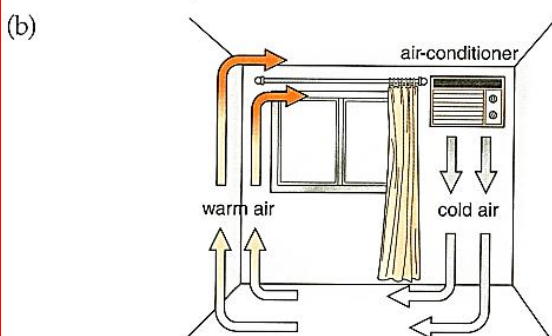
(a) Explain briefly why an air-conditioner is usually placed high on a wall. (3 marks)

(b) Draw arrows in the figure to show how warm air and cold air move in the room. Label the arrows. (2 marks)

(c) Closing the curtain can help cool the room. Explain briefly. (1 mark)

(d) Suggest another method to help cool the room. (1 mark)

(a) The cold air coming out from the air-conditioner sinks. 1A
This pushes the warm air up, which is taken in by the air-conditioner and cooled. 1A
This results in a convection current and the room can be cooled evenly. 1A

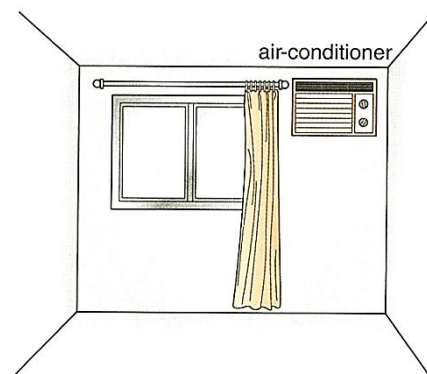


(Correct movement of air) 1A

(Correct labels of warm air and cold air) 1A

(c) This reduces the energy entering the room through the window by radiation. 1A

(d) Turn on a fan. 1A
(Or other reasonable answers)



違者必究

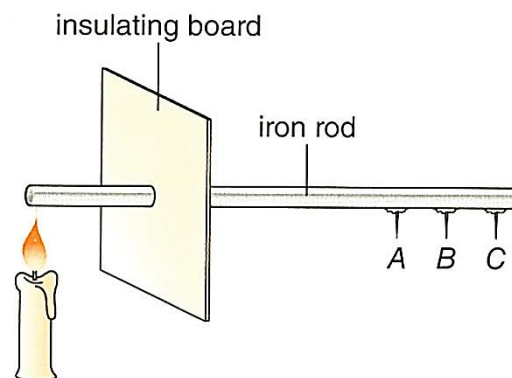
22. Three pins, A, B and C, are stuck on an iron rod at regular intervals by using the same amount of wax. The rod is heated as shown below.

- What happens to the pins? Explain briefly. (2 marks)
- What is the function of the insulating board? (1 mark)
- If a copper rod, instead of the iron rod, is used, what happens to the pins? (1 mark)
- Explain why the same amount of wax is used to stick the pins. (1 mark)

- | | |
|---|----------|
| (a) The pins drop one after another starting from pin A to pin C.
This is because heat is transferred from the hot part to the cold part of the rod by conduction. | 1A
1A |
| (b) The insulating board is used to avoid heat transfer due to radiation. | 1A |
| (c) The pins drop within a shorter time in the same order. | 1A |
| (d) This ensures that the same amount of energy is needed to melt each piece of wax. | 1A |

STRATEGY

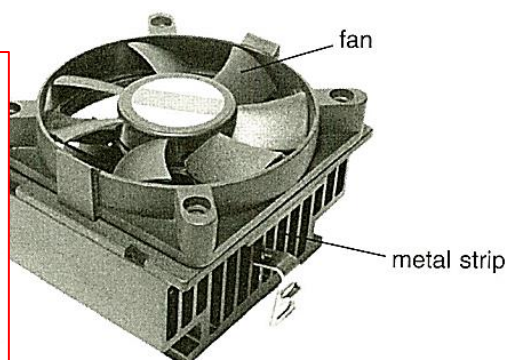
The wax holding B may melt first if it is smaller than that holding A. This affects the accuracy of the experiment.



23. The figure below shows the fan heat sink of the CPU processor in a computer. Metal strips are placed between the processor and the fan. The processor and the metal strips are in contact.

- Name the main process by which heat is transferred
 - from the processor to the metal strips; (1 mark)
 - from the metal strips to the surrounding air. (1 mark)
- Explain briefly the function of the fan. (2 marks)
- Explain the advantage of painting the metal strips black. (2 marks)
- Explain briefly what happens to the rate of change of the temperature of the processor if plastic strips are used instead of metal strips. (3 marks)
- Should the heat capacity of the metal strips be high or low? Explain briefly. (2 marks)

- | | |
|---|----------------|
| (a) (i) Conduction | 1A |
| (ii) Conduction | 1A |
| (b) The fan increases the air flow around the heat sink so that the warm air moves away from the heat sink and carries the heat away. | 1A
1A |
| (c) Black surfaces are good radiators of heat.
Heat can be transferred away from the metal strips by radiation effectively. | 1A
1A |
| (d) Plastic is a better insulator of heat than metal.
Hence, energy is transferred more slowly from the processor to the strips by conduction.
Therefore, the temperature of the processor decreases at a lower rate. | 1A
1A
1A |
| (e) The heat capacity of the metal strips should be high so that more energy is carried away from the processor for the same rise in temperature of the metal strips. | 1A
1A |



24. An architect plans to build a glass house immediately next to his home to make good use of solar energy in the winter (Fig a and b). There are 2 openings at the top and at the bottom of the wall common to his home and the glass house. They allow air to freely flow between the glass house and his home.

On a sunny day in the winter, the temperature inside the glass house can be much higher than that outside.

(a) Explain briefly how this happens. (6 marks)

(b) (i) On Figure b, use arrows to indicate the movement of air between the glass house and the architect's home.(1 mark)

(ii) Name this type of energy transfer. (1 mark)

(c) Which colour is the most suitable for the wall common to the glass house and his home? Explain briefly. (2 marks)

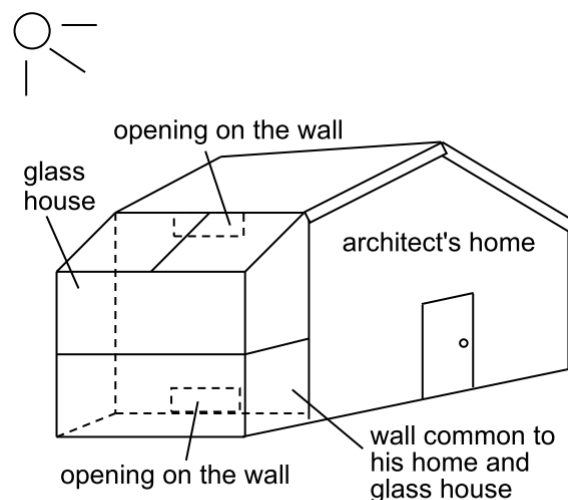


Figure a

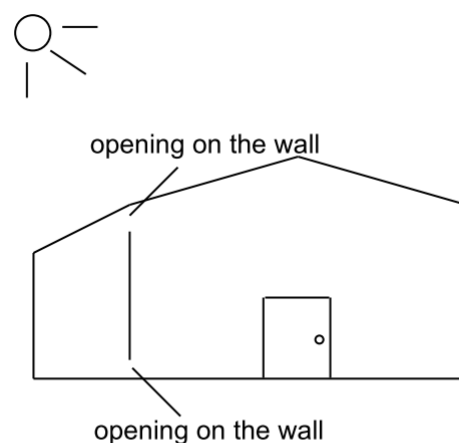
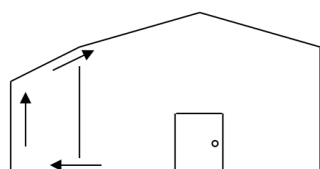


Figure b

(a) Solar radiation entering the glass house warms up the objects inside.	1A
These objects emit infra-red radiation,	1A
but this radiation cannot pass through the glass.	1A
Therefore, energy is trapped inside the glass house.	1A
In addition, the warm air is (partly) trapped inside the house.	1A

This greatly reduces the energy loss by convection.	1A
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(b) (i)	
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(Correct direction of air flow)

(ii) Convection

(c) Black is the most suitable	1A
because a black object is a good absorber of radiation.	1A